

Step size, floating-point precision and the predictability horizon of a simulated double pendulum: a student computational project

Luke Wang and Arjun Pemmasani

Harvey Mudd College, Claremont, CA 91711, United States of America

E-mail: apemmasani@hmc.edu

Abstract

We report a computational project, begun in an undergraduate course in numerical analysis, that asks for how long a computed trajectory is accurate. Students who simulate chaotic systems rarely ask this question about their own programs. We integrate the double pendulum with a fixed-step fourth-order Runge–Kutta method in arbitrary-precision arithmetic, so we can vary the step h and the number of significant bits p independently and measure the effects of truncation and of roundoff separately. The time T for which a run agrees with a far more accurate reference obeys a single law governed by the largest Lyapunov exponent λ . Each halving of h extends T by $4 \ln 2 / \lambda$ and each additional bit extends this time by $\ln 2 / \lambda$. For any given precision there is a best step beyond which refinement shortens T . The slopes yield $\lambda = 0.35 \pm 0.02$ and 0.35 ± 0.01 , which are equal to the growth rate of a perturbed trajectory over the same interval and within 20% of the long-time exponent 0.299 from the standard two-trajectory method. In double precision the pendulum cannot be followed for more than about 101 natural time units with this method, but its energy stays constant to a part in 10^{10} throughout. The project is suitable for advanced undergraduates in physics or applied mathematics and requires only a short program. It shows that numerical error is a perturbation like any other, and that its growth measures the dynamics.

Keywords: double pendulum, chaos, Lyapunov exponent, numerical integration, round-off error, computational physics education

1 Introduction

The double pendulum is among the simplest mechanical systems that are chaotic, and it is correspondingly popular in teaching. In this journal, D’Alessio [1] has given a com-

bined analytical, numerical, and experimental treatment, including the full spectrum of Lyapunov exponents, aimed at upper-year undergraduates. Earlier pedagogical studies measured the divergence of two real pendulums released together [2], developed the apparatus as a laboratory experiment [3], and mapped the onset of chaos numerically as a function of energy [4]. Calvão and Penna [5] reviewed in this journal the numerical methods used to study the system, and a recent pedagogical review is given in reference [6]. In all of them, the simulation is a tool, and its output is taken to be the motion of the pendulum, but this paper studies the tool.

Sensitive dependence on initial conditions means that a small perturbation grows exponentially, and a numerical integrator supplies a small perturbation at every step, by truncating the Taylor series of the solution and by rounding each arithmetic result. Hence, the computed trajectory must leave the true trajectory after a finite time, however carefully the program is written. Lorenz [7] met chaos in this way, and Lighthill [8] later called the time in question the horizon of predictability. The reliability of long computations of chaotic systems has since been examined in atmospheric modelling [9] and with multiple-precision arithmetic [10]. In particular, Li *et al* [11] showed that a given precision has a best step and a longest reliable time, and Kehlet and Logg [12] derived an error estimate of the form we use below. Wang and Li [13] found that the reliable time is linear in the number of digits, with a slope set by the Lyapunov exponent, and Mendes and Nepomuceno [14] estimated the exponent from the divergence of two equivalent computations that round differently. These studies treat the Lorenz and similar systems and are written for researchers. For students, Hayes [15] discussed what a computed chaotic trajectory means, Faux and Godolphin [16] collected floating-point pitfalls, and Wild [17] varied the word size in simulations of the double pendulum in an undergraduate thesis. However, in our experience, students who simulate a double pendulum are seldom asked how long their result can be believed, and the informal checks they apply, such as watching the energy, cannot answer this question.

This work began as our final project in an undergraduate course on numerical analysis. We simulated identical double pendulums with different steps and different floating-point formats and watched the trajectories separate. The observation that prompted this paper was that a step ten times smaller, at ten times the cost, added only five to ten seconds of agreement. Here we redo the project with a cleaner design, explain that observation, and show that the numerical error measures the Lyapunov exponent, so the explanation is quantitative.

We believe the project is useful to physics education at the advanced undergraduate level in three ways. First, it gives students a concrete meaning for the Lyapunov exponent in terms of something they control. Second, it separates truncation error from roundoff error, which are usually taught together and rarely distinguished in practice. Finally, it corrects two common beliefs, that a smaller step always gives a better answer and that

conservation of energy certifies a trajectory.

2 The project

2.1 Model

Two point masses m are suspended in series from a fixed pivot by massless rigid rods of length l and move in a vertical plane. The angles θ_1 and θ_2 are measured from the downward vertical. Measuring time in units of $\sqrt{l/g}$ (which is 0.319 s for $l = 1$ m) removes every parameter from the Lagrangian equations of motion, which become

$$\ddot{\theta}_1 = \frac{\omega_1^2 \sin \Delta \cos \Delta + \sin \theta_2 \cos \Delta + \omega_2^2 \sin \Delta - 2 \sin \theta_1}{2 - \cos^2 \Delta}, \quad (1)$$

$$\ddot{\theta}_2 = \frac{-\omega_2^2 \sin \Delta \cos \Delta + 2(\sin \theta_1 \cos \Delta - \omega_1^2 \sin \Delta - \sin \theta_2)}{2 - \cos^2 \Delta}, \quad (2)$$

with $\omega_i = \dot{\theta}_i$ and $\Delta = \theta_2 - \theta_1$. The conserved energy in units of mgl is

$$E = \omega_1^2 + \frac{1}{2}\omega_2^2 + \omega_1\omega_2 \cos \Delta - 2 \cos \theta_1 - \cos \theta_2. \quad (3)$$

The pendulum is released from rest with both rods horizontal, so $E = 0$.

2.2 Numerical method

Equations (1) and (2) are written as a first-order system for $\mathbf{y} = (\theta_1, \theta_2, \omega_1, \omega_2)$ and advanced with the classical fourth-order Runge–Kutta method (RK4) [18] at a fixed step h . We avoid adaptive step-size control deliberately, since its purpose is to hide the dependence on h that we want to see.

The arithmetic is done by the MPFR library [19], which rounds each operation correctly to a significand of p bits, where p is set by the program. Our first version used Julia’s `BigFloat` type and the present version uses Python with `gmpy2`, and both are thin wrappers around MPFR. With $p = 24, 53, 64$, and 113 the significand has the width of the IEEE single, double, extended, and quadruple formats [20]. With $p = 237$, the width of the 256-bit format, the rounding unit is 10^{-71} , so the arithmetic is exact for our purposes.

The choice of step proved important, since our first version used decimal steps such as $h = 10^{-4}$ and no binary format represents 10^{-4} exactly. Hence, runs at different precisions were also using slightly different steps. We now take $h = 2^{-k}$, which is exact at every precision, so changing p changes only the arithmetic.

2.3 Measuring error without an exact solution

We measure the error of a run by the distance $d(t)$, in units of l , between its lower bob and that of a reference run. Because both bobs lie within $2l$ of the pivot, d saturates near 4 when two runs have become unrelated. The *horizon* T of a run is the first time at which $d \geq 10^{-2}$, which is about when two superimposed animations can be told apart by eye.

The reference is a much more accurate run, and an elementary argument says how far the reference can be trusted. If e_a and e_r are the true errors of a run and of the reference, the triangle inequality gives $|d - e_a| \leq e_r$. Both errors are amplified by the same dynamics, so their ratio stays near its initial value, which for RK4 runs with steps 2^{-k_a} and 2^{-k_r} is $2^{-4(k_r - k_a)}$. With $k_r = 18$, the separation d equals the true error of every run with $k_a \leq 16$ to better than half a per cent, until that run reaches its own horizon.

2.4 Three experiments

Truncation only. With $p = 237$, roundoff is fifty orders of magnitude below truncation, so we vary k from 4 to 17 and compare with $k = 18$.

Roundoff only. With h fixed, we vary p from 11 to 128 and compare with the run at $p = 237$ and the *same* h . The two runs iterate the same discrete map, so their truncation errors are identical and cancel, which leaves only rounding. Ordinary hardware arithmetic offers no way of making this separation.

Both. With $p = 24, 32, 40, 53$, and 64 , we vary k and compare with the reference of the first experiment. This is the position of a student using single or double precision.

3 A model for the horizon

Let e_n be the error after n steps, and let J_n be the Jacobian of the flow over the next step. Then, $e_{n+1} = J_n e_n + \tau_n + \rho_n$, where $\tau_n = O(h^5)$ is the local truncation error and $\rho_n = O(2^{-p})$ is the rounding error. Thus, the error at time t is a sum of the errors introduced at earlier steps, each propagated by the product of the Jacobians in between. For a chaotic trajectory, that product grows on average as $e^{\lambda(t-t_j)}$, where λ is the largest Lyapunov exponent, so the sum is dominated by its earliest terms:

$$e(t) \approx \left(A h^4 + B 2^{-p} h^{-\gamma} \right) e^{\lambda t}. \quad (4)$$

The first term comes from $1/h$ truncation errors of size h^5 per unit time. The second comes from $1/h$ rounding errors of size 2^{-p} per unit time, where $\gamma = \frac{1}{2}$ if they accumulate as a random walk and $\gamma = 1$ if they accumulate systematically [21]. Kehlet and Logg [12]

derive $\gamma = \frac{1}{2}$. Setting $e(T) = \epsilon$ gives

$$T = \frac{1}{\lambda} \ln \frac{\epsilon}{A h^4 + B 2^{-p} h^{-\gamma}}. \quad (5)$$

With $h = 2^{-k}$ the two limiting cases are

$$T = T_A + \frac{4 \ln 2}{\lambda} k \quad (\text{truncation dominant}), \quad (6)$$

$$T = T_B + \frac{\ln 2}{\lambda} (p - \gamma k) \quad (\text{roundoff dominant}). \quad (7)$$

Halving the step extends the horizon by a fixed amount when truncation dominates and shortens it when roundoff dominates. Meanwhile, each extra bit extends the horizon by a fixed amount. The regimes meet at $k^* = [p + \log_2(A/B)]/(4 + \gamma)$, where the horizon takes its greatest value for that precision, $T_{\max} \approx \text{const} + (\ln 2/\lambda) 4p/(4 + \gamma)$. Note that these slopes depend on the pendulum only through λ .

4 Results

4.1 Step size

Figure 1 shows three pendulums that differ only in step. They are indistinguishable for tens of time units and then separate one at a time, largest step first. Figure 2(a) shows the corresponding errors, which rise at a common average rate over many orders of magnitude. They are evenly spaced because they are the same curve scaled by h^4 , as figure 2(b) shows. Multiplied by h^{-4} , they coincide to within 0.21 decades for every $h \leq 2^{-8}$ wherever none has saturated, and the error at $t = 10$ scales as $h^{4.000}$.

The common curve is the amplification factor of the trajectory. To check this, we perturbed $\theta_1(0)$ by 2^{-170} , which is small enough to stay in the linear regime for the whole run. The resulting separation, drawn as the thin line in figure 2(b), follows the collapsed error curves through every plateau and burst. The integrator's error grows as a perturbation of the initial condition grows because the error is itself a perturbation. The growth is exponential only on average, since the local rate varies as the pendulum visits more and less unstable configurations.

Figure 3(a) shows the horizon against k . A line fitted over $7 \leq k \leq 16$ has slope 7.9 ± 0.4 time units per halving, so equation (6) gives $\lambda = 0.35 \pm 0.02$. The scatter about the line is real, since a horizon that falls just before a burst of growth gains little from the next halving and a horizon that falls just after gains a great deal. The grey staircase in figure 3(a) is the horizon predicted for every h from the measured growth of the 2^{-170} perturbation, where the one constant A of equation (4) is fixed by the collapse. The staircase passes through the measured horizons step for step.

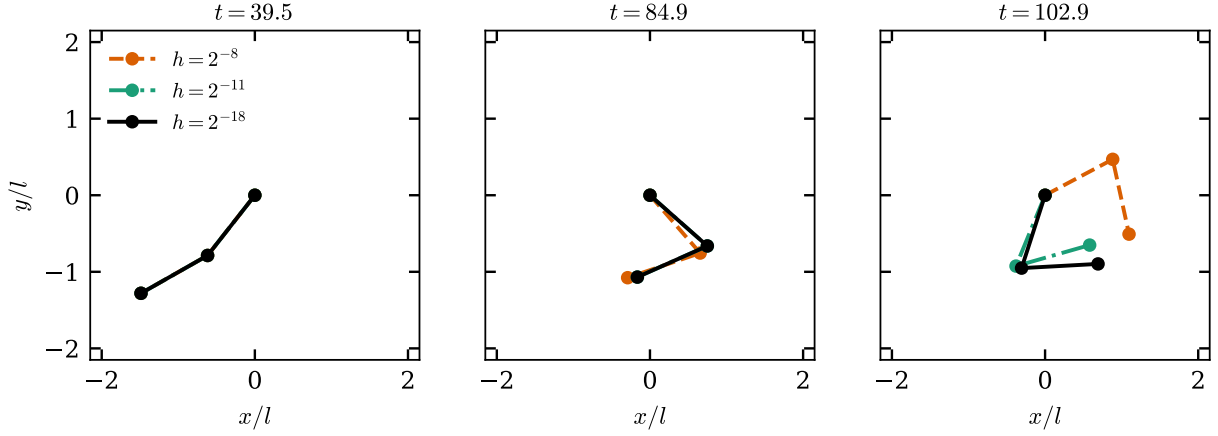


Figure 1: Three double pendulums released from the same state and integrated with the same 237-bit arithmetic, differing only in step h . The run with $h = 2^{-8}$ separates near $t = 79$ and the run with $h = 2^{-11}$ near $t = 97$ (time in units of $\sqrt{l/g}$). An animation is provided as supplementary material.

4.2 Precision

Figure 4 shows the second experiment at $h = 2^{-12}$, where the pattern is that of figure 2 with 2^{-p} in place of h^4 . The horizons in figure 3(b) have slope 1.99 ± 0.06 time units per bit, so equation (7) gives $\lambda = 0.35 \pm 0.01$. Changing the step by a factor of 256 shifts the line only slightly. Comparing errors at fixed early times for $h = 2^{-8}$ and 2^{-16} gives $\gamma \approx 0.6$, which is close to the random-walk value.

We left the lowest precisions out of the fit because with $p = 11$ and $h = 2^{-12}$ the pendulum never moves (the result we remember best from the project). At first the velocity ω_1 falls by exactly h per step for 2048 steps. At $t = \frac{1}{2}$ it reaches $-\frac{1}{2}$, where round-to-even freezes the velocity because h is half a unit in the last place of ω_1 . The increment to the angle is then $h|\omega_1| = 2^{-13}$ for ever, a quarter of the 2^{-11} needed to change the last of eleven bits in $\theta_1 = \pi/2$. Hence, the pendulum hangs horizontally with a constant velocity that produces no motion. With the same eleven bits and $h = 2^{-8}$ the increment first exceeds 2^{-11} at $t = 0.129$ and the pendulum falls, so reducing the step has made the error total. Since the velocity freezes at 2^{p-k} and the angle needs $|\omega_1| > 2^{k-p}$, the pattern is general for this release, and the pendulum stalled for every $h < 2^{-p}$ we tried and moved for every $h > 2^{-p}$. At $h = 2^{-p}$ the increment is itself a tie, so the outcome depends on the last bit of $\pi/2$.

4.3 The Lyapunov exponent four ways

Table 1 compares the estimates. The conventional value is from the two-trajectory method of Benettin *et al* [22–24]. A shadow orbit at distance 10^{-8} is advanced with the fiducial orbit, and once per time unit the logarithm of the growth of their separation is accumulated and the shadow is returned to its original distance. Over 2×10^4 time units this gives

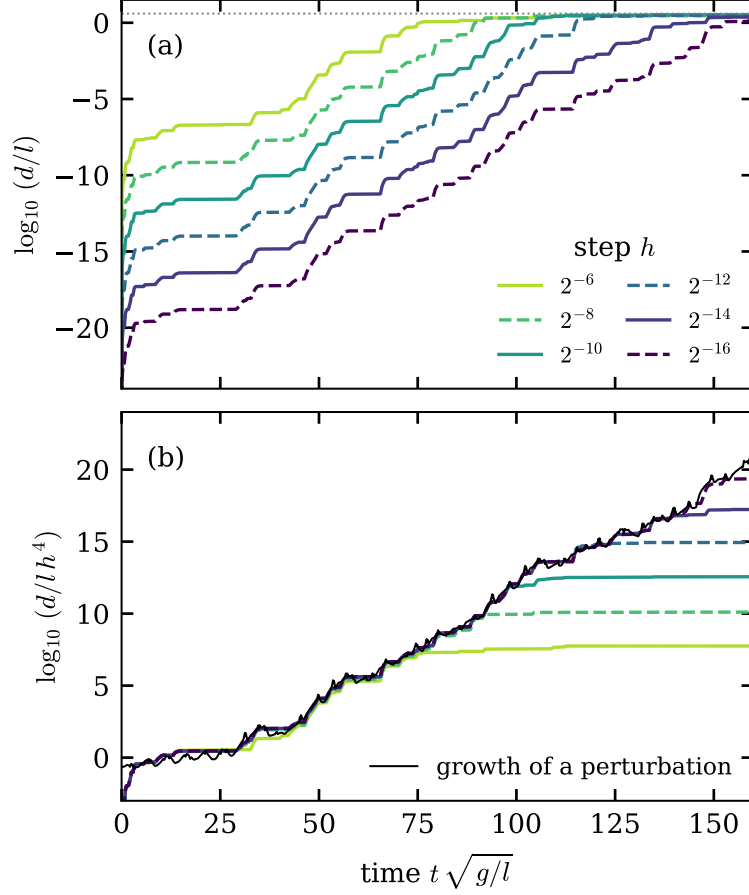


Figure 2: (a) Separation d of the lower bob from that of the reference ($h = 2^{-18}$) in 237-bit arithmetic (running maximum). The dotted line is the largest possible separation. (b) The same curves multiplied by h^{-4} . The thin black line is the separation of two runs whose initial conditions differ by 2^{-170} , shifted vertically.

$\lambda = 0.299$. Of 47 other states at rest with the same energy, 44 are chaotic with median 0.296, and 3 lie on regular orbits. The direct growth rate of the 2^{-170} perturbation is 0.33 over the whole run and 0.35 for $t > 20$, after the slow start from rest.

The two values obtained from numerical error agree with each other and with the direct growth rate over the same interval. All three exceed the long-time value by some 18%. This is not a failure of the method, since a horizon samples the growth rate only up to the horizon, and over its first 160 time units this trajectory is more unstable than on average. The exponent proper is a long-time limit, and finite-time values scatter about this limit. Students who picture exponential divergence as uniform do well to learn this distinction.

4.4 Working precision

Figure 5 shows the third experiment, where all precisions agree at large steps, since truncation dominates. As the step is reduced, each curve leaves the 237-bit line in turn and flattens, and at the lower precisions it then falls, as equation (5) predicts. Table 2

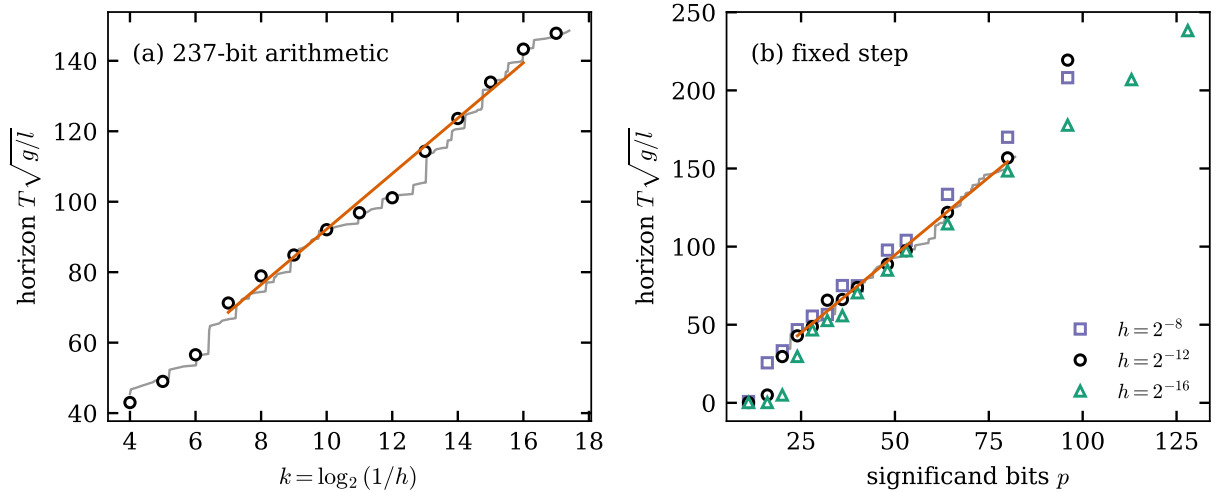


Figure 3: Horizon T , the first time at which $d \geq 10^{-2}$. (a) Against $k = \log_2(1/h)$ in 237-bit arithmetic (truncation only). (b) Against the number of significant bits p at fixed step (roundoff only). The straight lines are least-squares fits whose slopes give λ through equations (6) and (7). The grey staircases are the horizons predicted from the measured growth of a perturbation of the initial condition.

Table 1: Largest Lyapunov exponent for release from rest with both rods horizontal, in units of $\sqrt{g/l}$. The last three rows sample only the first 160 time units, over which the mean growth rate is 0.28 in the first half and 0.33 in the second. Uncertainties are standard errors of the fitted slopes. At $h = 2^{-8}$ and 2^{-16} the last row becomes 0.31 and 0.34.

Method	λ
Two-trajectory method, 2×10^4 time units	0.299
Growth of a 2^{-170} perturbation	0.33
Horizon against step, equation (6)	0.35 ± 0.02
Horizon against precision, equation (7)	0.35 ± 0.01

lists the best step and longest horizon for each precision. In double precision the longest horizon is 101 time units (32 s for one-metre rods), reached at $h = 2^{-11}$. Reducing the step further to 2^{-17} costs 64 times as much and gives 98. In single precision the best horizon is 53, at $h = 2^{-7}$, and by $h = 2^{-17}$ it has fallen to 30.

4.5 Energy

Figure 6 shows both errors for an ordinary double-precision run with $h = 2^{-10}$. The energy error drifts slowly upward, as expected of a non-symplectic method, and never exceeds 10^{-10} , but the trajectory is lost at $t \approx 92$. Energy confines the state to a three-dimensional surface in phase space, and chaotic divergence takes place within that surface, so a computed state can wander arbitrarily far without leaving the surface.

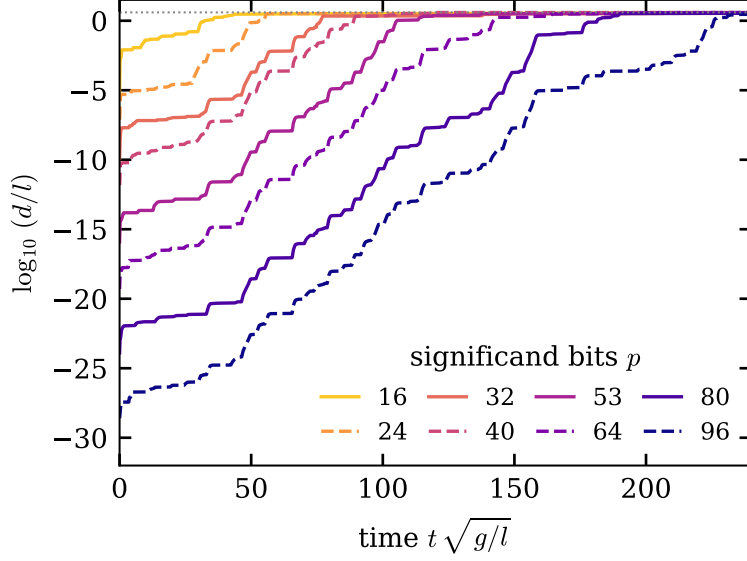


Figure 4: Separation from the 237-bit run at the same step, $h = 2^{-12}$, for eight significand widths. Truncation error is common to each pair and cancels, so only roundoff is seen.

Table 2: Longest horizon attainable with RK4 at each precision. The last column is for $l = 1$ m.

Significand bits p	Best step	$T_{\max}\sqrt{g/l}$	$T_{\max}$ (s)
24 (single)	2^{-7}	53	17
32	2^{-7}	66	21
40	2^{-9}	79	25
53 (double)	2^{-11}	101	32
64 (extended)	2^{-15}	124	40

5 Discussion

5.1 The price of a longer horizon

The work of a fixed-step run is proportional to 2^k . From equation (6), a gain of ΔT needs $\Delta k = \lambda \Delta T / (4 \ln 2)$, which multiplies the work by $e^{\lambda \Delta T / 4}$, and the precision must rise by about $\lambda \Delta T (4 + \gamma) / (4 \ln 2)$ bits to stay at the optimum. Doubling the double-precision horizon of this pendulum means extrapolating beyond the times sampled, so we use the long-time exponent 0.299. The number of steps is multiplied by about 2000 and some 50 more bits are needed. At the finite-time rate 0.35 of the first 160 time units the figures would be nearer 7000 and 59, but either way the horizon grows as the logarithm of the effort. In everyday units the long-time exponent is $\lambda = 0.94 \text{ s}^{-1}$ for one-metre rods, so reducing the step tenfold adds $4 \ln 10 / \lambda \approx 10 \text{ s}$ (8 s at the finite-time rate), as we saw in our first crude runs.

A method of order n replaces 4 by n , and we verified that the error of forward Euler at fixed time scales as $h^{1.002}$ and that of the explicit midpoint rule as $h^{1.999}$. Forward Euler

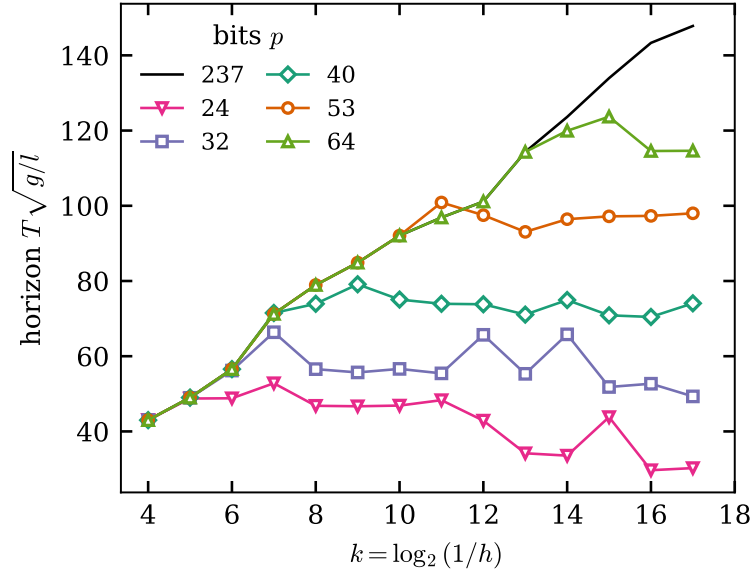


Figure 5: Horizon against step for five working precisions, with the 237-bit result for comparison. Each precision has a step beyond which refinement no longer helps.

with $h = 2^{-16}$ has a horizon of 31 time units, shorter than the 43 of RK4 with $h = 2^{-4}$, so for chaotic problems the case for high-order methods is stronger than usual.

5.2 What survives the horizon

After T the computed trajectory is no longer the solution of the problem posed. However, shadowing results [15, 25–28] show that for long times it stays close to an exact trajectory with slightly different initial data. Hence, averages over the motion are still computed correctly, and the first row of table 1 relies on this fact. As a check, we computed the time-averaged kinetic energy over 10^4 time units in single and double precision. The two trajectories part after about fifty time units, but the averages are 1.357 and 1.358.

5.3 Running the project with a class

The program is the usual RK4 with the number type changed, and apart from the two finest reference runs everything reported here finishes in minutes on a laptop. In the natural sequence we followed, students first overlay animations of runs that differ in step or precision and see them separate. They then plot the separation on a logarithmic scale and find straight lines, fit the horizons, and compare the resulting λ with a two-trajectory estimate. Students without a multiple-precision library can still carry out the third experiment, using double precision at the best step of table 2 as the reference for single precision. A natural extension is other energies, where λ is smaller and some orbits are regular, so the error grows only as a power of t . Another is a symplectic integrator, to find out whether preserving phase-space structure helps with prediction. A third is other

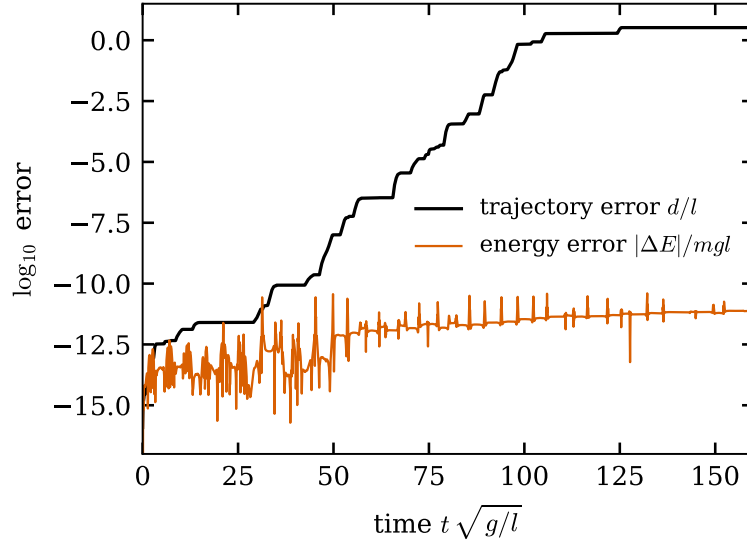


Figure 6: Error in the position of the lower bob and in the energy for a run in double precision with $h = 2^{-10}$.

chaotic systems, such as the Lorenz equations, for which the double-precision horizon can be predicted from the known exponent before it is measured.

6 Conclusion

A simulation of a chaotic system is accurate for a time that depends logarithmically on the error introduced per step. For the double pendulum integrated by RK4, that time increases by $4 \ln 2 / \lambda$ per halving of the step and by $\ln 2 / \lambda$ per bit of precision, up to a ceiling set by whichever source of error is larger. In double precision, the ceiling is about 101 natural time units. The same measurements give the finite-time Lyapunov exponent of the trajectory to within a few per cent, and its long-time value to within twenty per cent. Students who complete the project see that the system responds to truncation and roundoff as it responds to any other perturbation, and they learn to ask for how long any simulation is accurate.

Acknowledgments

This work began as a final project in Math 165, Numerical Analysis, at Harvey Mudd College.

Data availability statement

The code that produced every result and figure, the numerical data plotted, and two animations are provided as supplementary material.

Conflict of interest

The authors declare no conflict of interest.

References

- [1] Serge D'Alessio. An analytical, numerical and experimental study of the double pendulum. *Eur. J. Phys.*, 44(1):015002, 2023. doi: 10.1088/1361-6404/ac986b.
- [2] Troy Shinbrot, Celso Grebogi, Jack Wisdom, and James A. Yorke. Chaos in a double pendulum. *Am. J. Phys.*, 60(6):491–499, 1992. doi: 10.1119/1.16860.
- [3] R. B. Levien and S. M. Tan. Double pendulum: An experiment in chaos. *Am. J. Phys.*, 61(11):1038–1044, 1993. doi: 10.1119/1.17335.
- [4] Tomasz Stachowiak and Toshio Okada. A numerical analysis of chaos in the double pendulum. *Chaos Solitons Fractals*, 29(2):417–422, 2006. doi: 10.1016/j.chaos.2005.08.032.
- [5] A. M. Calvão and T. J. P. Penna. The double pendulum: a numerical study. *Eur. J. Phys.*, 36(4):045018, 2015. doi: 10.1088/0143-0807/36/4/045018.
- [6] Samuel Rocha de Oliveira. Deterministic chaos: A pedagogical review of the double pendulum case. *Rev. Bras. Ensino Fís.*, 46:e20240060, 2024. doi: 10.1590/1806-9126-RBEF-2024-0060.
- [7] Edward N. Lorenz. Deterministic nonperiodic flow. *J. Atmos. Sci.*, 20(2):130–141, 1963. doi: 10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2.
- [8] James Lighthill. The recently recognized failure of predictability in Newtonian dynamics. *Proc. R. Soc. Lond. A*, 407(1832):35–50, 1986. doi: 10.1098/rspa.1986.0082.
- [9] João Teixeira, Carolyn A. Reynolds, and Kevin Judd. Time step sensitivity of non-linear atmospheric models: Numerical convergence, truncation error growth, and ensemble design. *J. Atmos. Sci.*, 64(1):175–189, 2007. doi: 10.1175/JAS3824.1.
- [10] Shijun Liao. On the reliability of computed chaotic solutions of non-linear differential equations. *Tellus A*, 61(4):550–564, 2009. doi: 10.1111/j.1600-0870.2009.00402.x.
- [11] Jianping Li, Qingcun Zeng, and Jifan Chou. Computational uncertainty principle in nonlinear ordinary differential equations. II. Theoretical analysis. *Sci. China Ser. E*, 44(1):55–74, 2001. doi: 10.1007/BF02916726.

- [12] Benjamin Kehlet and Anders Logg. A posteriori error analysis of round-off errors in the numerical solution of ordinary differential equations. *Numer. Algorithms*, 76(1): 191–210, 2017. doi: 10.1007/s11075-016-0250-4.
- [13] PengFei Wang and JianPing Li. On the relation between reliable computation time, float-point precision and the Lyapunov exponent in chaotic systems, 2014. arXiv:1410.4919 [nlin.CD].
- [14] Eduardo M. A. M. Mendes and Erivelton G. Nepomuceno. A very simple method to calculate the (positive) largest Lyapunov exponent using interval extensions. *Int. J. Bifurcation Chaos*, 26(13):1650226, 2016. doi: 10.1142/S0218127416502266.
- [15] Wayne B. Hayes. Computer simulations, exact trajectories, and the gravitational N-body problem. *Am. J. Phys.*, 72(9):1251–1257, 2004. doi: 10.1119/1.1764561.
- [16] David A. Faux and Janet Godolphin. The floating point: Tales of the unexpected. *Am. J. Phys.*, 89(8):806–814, 2021. doi: 10.1119/10.0003915.
- [17] Rebecca Wild. The effects of finite precision on the simulation of the double pendulum. Senior Honors Project, James Madison University, 2019. <https://commons.lib.jmu.edu/honors201019/730>.
- [18] Ernst Hairer, Syvert P. Nørsett, and Gerhard Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*, volume 8 of *Springer Series in Computational Mathematics*. Springer, Berlin, 2nd rev. edition, 1993. doi: 10.1007/978-3-540-78862-1.
- [19] Laurent Fousse, Guillaume Hanrot, Vincent Lefèvre, Patrick Pélicier, and Paul Zimmermann. MPFR: A multiple-precision binary floating-point library with correct rounding. *ACM Trans. Math. Softw.*, 33(2):13, 2007. doi: 10.1145/1236463.1236468.
- [20] David Goldberg. What every computer scientist should know about floating-point arithmetic. *ACM Comput. Surv.*, 23(1):5–48, 1991. doi: 10.1145/103162.103163.
- [21] Ernst Hairer, Robert I. McLachlan, and Alain Razakarivony. Achieving Brouwer’s law with implicit Runge–Kutta methods. *BIT Numer. Math.*, 48(2):231–243, 2008. doi: 10.1007/s10543-008-0170-3.
- [22] Giancarlo Benettin, Luigi Galgani, Antonio Giorgilli, and Jean-Marie Strelcyn. Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems; a method for computing all of them. Part 1: Theory. *Meccanica*, 15(1): 9–20, 1980. doi: 10.1007/BF02128236.
- [23] Giancarlo Benettin, Luigi Galgani, Antonio Giorgilli, and Jean-Marie Strelcyn. Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian

- systems; a method for computing all of them. Part 2: Numerical application. *Mechanica*, 15(1):21–30, 1980. doi: 10.1007/BF02128237.
- [24] Alan Wolf, Jack B. Swift, Harry L. Swinney, and John A. Vastano. Determining Lyapunov exponents from a time series. *Physica D*, 16(3):285–317, 1985. doi: 10.1016/0167-2789(85)90011-9.
- [25] Stephen M. Hammel, James A. Yorke, and Celso Grebogi. Do numerical orbits of chaotic dynamical processes represent true orbits? *J. Complexity*, 3(2):136–145, 1987. doi: 10.1016/0885-064X(87)90024-0.
- [26] Stephen M. Hammel, James A. Yorke, and Celso Grebogi. Numerical orbits of chaotic processes represent true orbits. *Bull. Am. Math. Soc.*, 19(2):465–469, 1988. doi: 10.1090/S0273-0979-1988-15701-1.
- [27] Celso Grebogi, Stephen M. Hammel, James A. Yorke, and Tim Sauer. Shadowing of physical trajectories in chaotic dynamics: Containment and refinement. *Phys. Rev. Lett.*, 65(13):1527–1530, 1990. doi: 10.1103/PhysRevLett.65.1527.
- [28] Tim Sauer, Celso Grebogi, and James A. Yorke. How long do numerical chaotic solutions remain valid? *Phys. Rev. Lett.*, 79(1):59–62, 1997. doi: 10.1103/PhysRevLett.79.59.